



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
ISE-I (Semester-I) 2025-26

Class: First Year M. Tech.

Branch: All branches

Course: Open Elective: Operations Research

Course Code: OEC-803

Date: 17-Oct-2025

Time: 16:30 to 17:15

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	Define systems engineering. Differentiate between systems approach and system analysis. How is it related to operations research?	07	1	1
Q. 2)	What is a model? Explain in brief role of systems models and state the different types of system models.	07	3	2
Q. 3)	Explain the concept of system with a neat diagram.	06	4	2

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Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science and Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination I

Class: MTech CNIS First Year

Information Security

Course Code: PCC-NS517

Date: 13/10/2025

Time: 4.30 to 5.15

Max. Marks: 20

Note:

- Solve any two questions.
- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Apply the CIA triad to a hospital information system.	5	CO4	BT 3
	b	Evaluate the importance of integrating security in the SDLC.	5	CO5	BT 5
Q.2	a	Why is the top-down approach to information security superior to the bottom-up approach?	5	CO4	BT 4
	b	What is information extortion? Describe how such an attack can cause losses.	5	CO3	BT 3
Q.3	a	I. Explain the difference between a brute-force and a dictionary attack on passwords. II. How can multi-factor authentication (MFA) mitigate the risks of password-based attacks?	5	CO3	BT 3
				CO4	BT4
	b	Why is DDoS generally harder to defend against than a single-source DoS attack?	5	CO3	BT 3

== ALL THE BEST ==

CO1 :Define the concepts of and definition of information system
CO2:Learn several types of information system.
CO3:Apply threats to information security
CO4:Analyze how to protect information resources
CO5:Design and develop maintaining and protecting information system.

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create



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